

HLAD 7109: Healthcare Informatics

Assignment 1

Total Points: 10 (5 questions × 2 points)

Submission Format: Submit **Word (.docx)** or **PDF** via the course LMS.

Instructions: Answer each question in **3–6 sentences**. You can use LLMs without copy-pasting.

Questions

Q1. (2 points) Explain the difference between *data*, *information*, and *knowledge*. Define each term and provide **one healthcare example** for each (data, information, knowledge).

Q2. (2 points) In the Data-to-Knowledge pyramid, explain **how context changes data into information**. Use a short example (e.g., a lab value such as blood glucose) to show why context matters in healthcare.

Q3. (2 points) Define **Protected Health Information (PHI)** and explain why PHI requires safeguards. Then, briefly describe how a **Personal Health Record (PHR)** can include PHI.

Q4. (2 points) List and explain **two consequences of poor data quality** in healthcare (e.g., impact on patient safety, costs, privacy/security, liability). For each consequence, provide a 1–2 sentence example scenario.

Q5. (2 points) Describe common **data quality challenges** (e.g., missing, inconsistent, erroneous data; workflow mismatch; human factors). Choose **one challenge** and propose **one practical solution** using health IT.

Academic Integrity Note

All work must be original and written in your own words. Properly cite any external sources if used.